

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration through Form
FR-1	User Registration & Authentication	Login using Email and Password
FR-1	User Registration & Authentication	JWT-based User Authentication
FR-2	User Profile Management	View Profile Information
FR-2	User Profile Management	Update Profile Details
FR-2	User Profile Management	Manage Delivery Address
FR-3	Product Management	View All Products
FR-3	Product Management	Search Products
FR-3	Product Management	Filter Products by Category
FR-3	Product Management	View Product Details
FR-4	Cart Management	Add Products to Cart
FR-4	Cart Management	Update Product Quantity
FR-4	Cart Management	Remove Products from Cart
FR-4	Cart Management	View Cart Summary
FR-5	Order Management	Place Orders
FR-5	Order Management	View Order History
FR-5	Order Management	Track Order Status
FR-6	Admin Management	Add Products
FR-6	Admin Management	Edit Products

FR-6	Admin Management	Delete Products
FR-6	Admin Management	Manage Customer Orders
FR-6	Admin Management	Manage Registered Users
FR-7	Banner Management	Add Promotional Banners
FR-7	Banner Management	Delete Promotional Banners

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system provides an easy-to-use and responsive user interface for customers and administrators.
NFR-2	Security	User passwords are encrypted and JWT authentication is used to secure access to the application.
NFR-3	Reliability	The application ensures consistent operation with proper error handling and data validation.
NFR-4	Performance	Product search, login, and order processing operations respond within acceptable time limits.
NFR-5	Availability	The application remains accessible whenever the server and database services are operational.
NFR-6	Scalability	The system can support increasing numbers of users, products and issues